

Sicheng Ouyang

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Education

University of Waterloo, Waterloo, ON

Sep 2025 – Present

Bachelor of Software Engineering, Expected Apr 2030

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Frameworks & Libraries: FastAPI, React, Next.js, PyTorch

Data & Infrastructure: PostgreSQL, PostGIS, MySQL, Redis, Supabase, Docker, Linux, GitHub Actions

Experience

Mui Scientific

SQL · REST APIs · ERP Integrations · Next.js

May 2026 – Present

Software Engineer

Mississauga, ON

- Implemented SQL/REST synchronization workflows between enterprise resource planning and inventory systems, enabling serial/batch traceability across **10,000+** inventory records and reducing audit preparation from roughly half a day to **30 minutes**.
- Automated cross-system reconciliation for procurement, warehouse, and sales records using validation rules and exception reporting, eliminating manual review for roughly **80%** of routine inventory checks.
- Migrated the corporate site from PHP/WordPress to Next.js, consolidating duplicated page templates into reusable components and standardizing the deployment workflow.

KnowYi Technology Inc.

Python · Next.js · OpenAI API · Async Pipelines

Apr 2025 – May 2025

Software Engineer, Founding Team

Waterloo, ON

- Engineered OpenAI-assisted extraction pipelines for PDF, Excel, and native CAD files, generating candidate values from drawing tables before AI parsing and preserving source references for human review.
- Parallelized website research with asynchronous Python workflows, delivering one-click analysis in under **15 seconds**.

Tencent Music Entertainment Group

Python · Speech Synthesis · Audio ML

Jul 2024 – Aug 2024

Machine Learning Engineer

Shenzhen, China

- Evaluated 10 speech-generation models across 20+ controlled experiments, testing model combinations and tuning inference parameters for speaker consistency.
- Built reusable Mel-spectrogram and F0 preprocessing workflows to standardize cross-model evaluation.
- Improved an internal speaker-consistency metric by 20% and increased MOS from **3.43** to **3.63** through iterative model and inference tuning.

Advanced A-Level Study Website

WordPress · PHP · MySQL · Tencent Cloud

Sep 2023 – Aug 2025

Co-Founder & Lead Developer

Wuhan, China

- Co-founded and operated an A-Level learning platform reaching **1,000+** peak daily visitors and supporting **200+** students, owning WordPress/PHP development, MySQL operations, and Tencent Cloud deployment.

Projects

The Cobalt Guide

Next.js · TypeScript · Supabase/PostGIS · MapLibre · Redis

[Demo](#) [GitHub](#)

- Launched a Canada-wide platform helping users discover nearby credit-card rewards through PostGIS geospatial search and MapLibre maps.
- Hardened delivery with Supabase Auth/RLS, Redis rate limits, and a four-layer GitHub Actions pipeline spanning linting, unit, E2E, and RLS tests.

Real-Time Chat Platform

FastAPI · Next.js · WebSocket · PostgreSQL

[GitHub](#)

- Designed incremental WebSocket synchronization with paginated recovery, allowing reconnecting clients to retrieve missed persisted messages.
- Implemented JWT WebSocket authentication, SMTP verification, rate limits, and structured API errors to block unverified access.

Neural Network Playground

FastAPI · PyTorch · React · WebSocket

[Demo](#) [GitHub](#)

- Built an interactive neural-network playground with FastAPI, supporting configurable training and evaluation across **five** datasets.
- Streamed epoch-level metrics over WebSockets to React charts for live loss and training-status visualization.